Dynamical Oceanography

Problem Set 1

Due Monday 15 February, 2010

1. Briefly and informally derive the Boussinesq equations, beginning with the Navier Stokes equations, stating clearly any assumptions made.

2. Briefly and informally derive the planetary geostrophic equations, for (a) the shallow water equations and (b) the Boussinesq equations, stating clearly any assumptions made.

3. Consider the planetary geostrophic equations, on the f-plane. (That is, the Coriolis parameter is constant.)

   (a) Is this a realistic situation in either the Earth's atmosphere or ocean? If so, give an example in which the regime is valid, and if not, dream up a situation in which it might be valid, if possible.

   (b) Show that in the shallow water approximation, the advective term identically vanishes. What occurs in the fully stratified case? Discuss what this means physically, and how a real fluid might evolve under such circumstances.

4. Derive and discuss the formulation of the Stommel model for a quasi-geostrophic system. Compare this to the planetary geostrophic formulation, in terms of the scales and general validity of the two approaches.

5. Consider the steady nonlinear Stommel model in the form

$$J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = \nabla \times \tau - r\zeta, \tag{1}$$

   where the notation is standard. Show that the nonlinear term will be small in the interior of the domain if $\beta L^2/U \gg 1$ where L is the scale of the large-scale motion and U a representative velocity. Show (e.g., by using using scale analysis) that for sufficiently strong wind, the nonlinear term must become more important. Derive a criterion (in terms of τ and the other parameters in the equation) for this.